

UNIT4

2MARKS QUESTIONS

1. Define Machine Learning.

Machine learning is a subfield of computer science that evolved from the study of pattern recognition and computational learning theory in Artificial Intelligence (AI).

2. Define Learning.

A computer program which learns from experience is called a machine learning program or simply a learning program. Such a program is sometimes also referred to as a learner.

3. Give any two examples for learning problems.

- Handwriting recognition learning problem
- A robot driving learning problem

4. List the different components of learning.

- Data storage
- Abstraction
- Generalization
- Evaluation

5. Give any four applications of Machine learning.

- Recommendation System
- Spam detection
- Prospect customer identification
- In medicine, learning programs are used for medical diagnosis

6. List the different forms of machine learning.

- Supervised Learning
- Unsupervised Learning
- Semi-supervised machine learning
- Reinforcement Learning

7. What is supervised learning?

.Supervised learning is the machine learning task of learning a function that maps an input to an output based on example input-output pairs.

8. What is unsupervised learning?

Unsupervised learning is a type of machine learning algorithm used to draw inferences from datasets consisting of input data without labeled responses

9. What is semi-supervised learning?

This algorithms fall somewhere in between supervised and unsupervised learning, since they use both labeled and unlabeled data for training – typically a small amount of labeled data and a large amount of unlabeled data

10.What is reinforcement learning?

The basic objective of reinforcement learning algorithms is to map situations to actions that yield the maximum final reward. While mapping the action, the algorithm should not just consider the immediate reward but also next and all subsequent rewards

11.List the types of decision trees.

1. Classification trees (Yes/No types)
2. Regression trees (Continuous data types)

12.Define Entropy and Information Gain.

Entropy Entropy,also called as Shannon Entropy is denoted by $H(S)$ for a finite set S , is the measure of the amount of uncertainty or randomness in data.

Information Gain Information gain is also called as Kullback-Leibler divergence denoted by $IG(S,A)$ for a set S is the effective change in entropy after deciding on a particular attribute A . It measures the relative change in entropy with respect to the independent variables

13.What is Regression analysis? Give example.

Regression analysis is a statistical method to model the relationship between a dependent (target) and independent (predictor) variables with one or more independent variables

Some examples of regression can be as:

- Prediction of rain using temperature and other factors
- Determining Market trends
- Prediction of road accidents due to rash driving.

14.List any four terminologies related to regression analysis.

- Dependent Variable
- Independent Variable
- Outliers.
- Multicollinearity
- Underfitting and Overfitting

15.List the different types of Regression.

- Linear Regression
- Logistic Regression
- Polynomial Regression
- Support Vector Regression
- Decision Tree Regression
- Random Forest Regression
- Ridge Regression
- Lasso Regression

16.What do you mean by Simple linear regression and multiple linear regression?

- If there is only one input variable (x), then such linear regression is called **simple linear regression**. And if there is more than one input variable, then such linear regression is called **multiple linear regression**.

17.What do you mean by linear regression?

- Linear regression is a statistical regression method which is used for predictive analysis.
- It is one of the very simple and easy algorithms which works on regression and shows the relationship between the continuous variables.

18.What do you mean by logistic regression?

- Logistic regression is another supervised learning algorithm which is used to solve the classification problems. In **classification problems**, we have dependent variables in a binary or discrete format such as 0 or 1.

19.List the types of logistic regression.

- Binary(0/ 1, pass/fail)
- Multi(cats, dogs, lions)
- Ordinal(low, medium, high)

20.What is Artificial Neural Network?

An artificial neural network (ANN) is like a computing system that simulates how the human brain analyzes information and processes it. It is the branch of artificial intelligence (AI) and solves problems that may be difficult or impossible to solve such issues for humans

21.Write any four advantages of Artificial Neural Network. (SEE 6M)

22. Write any four disadvantages of Artificial Neural Network (SEE 6M)

23. List the types of Artificial Neural Network. (SEE 6M)

24. What do you mean by Autoencoders?

As the name suggests, an autoencoder aims to learn encoding as a representation of training sample data automatically without human intervention. The autoencoder is widely used for dimensionality reduction and data de-noising.

25. List the different types of layers in Convolution Neural Network.

CNN consists of four main types of layers: input layer, convolution layer, pooling layer, fully connected layer.

26. List the components of LSTM.

- Input gate layer
- Forget gate layer
- Output gate layer
- Memory cell state vector

27. List the types of SVM.

- Linear SVM
- Non-linear SVM

28. What is Hyperplane?

There can be multiple lines/decision boundaries to segregate the classes in n-dimensional space, but we need to find out the best decision boundary that helps to classify the data points. This best boundary is known as the hyperplane of SVM.

29. What do you mean by Support Vector?

The data points or vectors that are the closest to the hyperplane and which affect the position of the hyperplane are termed as Support Vector. Since these vectors support the hyperplane, hence called a Support vector.

30. List the types of Unsupervised learning algorithm.

- Clustering.
- Association

31. List any four Unsupervised learning algorithms.

- K-means clustering
- KNN (k-nearest neighbors)
- Hierarchical clustering
- Anomaly detection
- Neural Networks
- Apriori algorithm

32. Write any two advantages of Unsupervised learning.

- Unsupervised learning is used for more complex tasks as compared to supervised learning because, in unsupervised learning, we don't have labeled input data.
- Unsupervised learning is preferable as it is easy to get unlabeled data in comparison to labeled data.

33. Write any two disadvantages of Unsupervised learning.

- Unsupervised learning is intrinsically more difficult than supervised learning as it does not have corresponding output.
- The result of the unsupervised learning algorithm might be less accurate as input data is not labeled, and algorithms do not know the exact output in advance.

34. Differentiate Supervised learning and Unsupervised learning.

SEE 6M QS

35. Expand NLP, NER, NLU, NLG.

- Natural Language Understanding (NLU)
- Natural Language Generation (NLG)
- Named Entity Recognition (NER),
- NLP - Natural Language Processing,

36. Give any four applications of Artificial Intelligence.

NLP, Text Classification, Information Retrieval, Speech recognition, Image Processing And Computer Vision, Robotics

37. List the components of NLP.

- Natural Language Understanding (NLU)
- Natural Language Generation (NLG)

38. List the different phases of NLP.

- Lexical Analysis and Morphological
- Syntactic Analysis (Parsing)
- Semantic Analysis
- Discourse Integration
- Pragmatic Analysis

39. Give any four applications of NLP.

- Question Answering
- Spam Detection
- Sentiment Analysis
- Spelling correction
- Speech Recognition

40. List the NLP preprocessing techniques.

- Tokenization
- Bag-of-words models
- Stop word removal
- Stemming and lemmatization
- Part-of-speech tagging and syntactic parsing

41. Mention the different Use cases included in Text Classification.

- Sentiment Analysis
- Document Classification

- Spam Detection
- Intent Classification
- News Categorization

42.What do you mean by information retrieval?

Information retrieval is defined as a completely automated procedure that answers to a user query by reviewing a group of documents and producing a sorted document list that ought to be relevant to the user's query criteria.

43.List the types of Information Retrieval Model.

1. Classic IR Model
2. Non-Classic IR Model
3. Alternative IR Model

44.What do you mean by Speech Recognition?

Speech recognition, also called automatic speech recognition (ASR), computer speech recognition, or speech-to-text, is a form of artificial intelligence and refers to the ability of a computer or machine to interpret spoken words and translate them into text

45.Give any four applications of Image processing.

- Medical Imaging
- Surveillance and Security
- Industrial Inspection
- Scientific Image Analysis
- Photo Editing

46.What do you mean by Computer Vision?

Computer vision is a field of artificial intelligence that enables computers to derive meaningful information from digital images, videos, and other visual inputs

47.Differentiate Image processing and Computer Vision.(SEE 6M QUES)

48.List the types of Automation in Robotics.

- Fixed Automation
- Programmable Automation

- Flexible Automation

49. List the different types of Robots.

- Androids
- Telechir
- Telepresence robot .
- Industrial robot
- Swarm robot
- Smart robot

Long Answer Questions (3,4,5,6 Marks)

1. List and explain the different components of Machine learning.

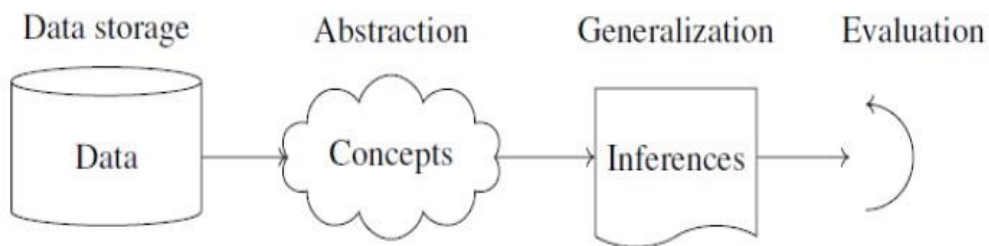


Figure 1.1: Components of learning process

1. Data storage

Facilities for storing and retrieving huge amounts of data are an important component of the learning process. Humans and computers alike utilize data storage as a foundation for advanced reasoning.

- In a human being, the data is stored in the brain and data is retrieved using electrochemical signals.
- Computers use hard disk drives, flash memory, random access memory and similar devices to store data and use cables and other technology to retrieve data.

2. Abstraction

The second component of the learning process is known as abstraction. Abstraction is the process of extracting

knowledge about stored data. This involves creating general concepts about the data as a whole. The creation of knowledge involves application of known models and creation of new models.

The process of fitting a model to a dataset is known as training. When the model has been trained, the data is transformed into an abstract form that summarizes the original information.

2. *Generalization*

The third component of the learning process is known as generalisation. The term generalization describes the process of turning the knowledge about stored data into a form that can be utilized for future action. These actions are to be carried out on tasks that are similar, but not identical, to those what have been seen before. In generalization, the goal is to discover those properties of the data that will be most relevant to future tasks.

3. *Evaluation*

Evaluation is the last component of the learning process. It is the process of giving feedback to the user to measure the utility of the learned knowledge. This feedback is then utilized to effect improvements in the whole learning process.

2. List and explain the applications of machine learning.

- In retail business, machine learning is used to study consumer behaviour.
- In finance, banks analyze their past data to build models to use in credit applications, fraud detection, and the stock market.

- In manufacturing, learning models are used for optimization, control, and troubleshooting.
- In medicine, learning programs are used for medical diagnosis.
- In telecommunications, call patterns are analyzed for network optimization and maximizing the quality of service.
- In science, large amounts of data in physics, astronomy, and biology can only be analyzed fast enough by computers. The World Wide Web is huge; it is constantly growing and searching for relevant information cannot be done manually.
- In artificial intelligence, it is used to teach a system to learn and adapt to changes so that the system designer need not foresee and provide solutions for all possible situations.
- It is used to find solutions to many problems in vision, speech recognition, and robotics.
- Machine learning methods are applied in the design of computer-controlled vehicles to steer correctly when driving on a variety of roads.
- Machine learning methods have been used to develop programmes for playing games such as chess, backgammon and Go.

3. List and explain the different forms of machine learning.

(ans 4,5,6)

4. Write a note on Supervised learning.

A training set of examples with the correct responses (targets) is provided and, based on this training set, the algorithm generalizes to respond correctly to all possible inputs. This is also called learning from exemplars. Supervised learning is the machine learning task of learning a function that maps an input to an output based on example input-output pairs.

In supervised learning, each example in the training set is a pair consisting of an input object (typically a vector) and an output value. A supervised learning algorithm analyzes the training data and produces a function, which can be used for mapping new examples. In the optimal case, the function will correctly determine the class labels for unseen instances. Both classification and regression problems are supervised learning problems. A wide range of supervised learning algorithms are available, each with its strengths and weaknesses. There is no single learning algorithm that works best on all supervised learning problems.

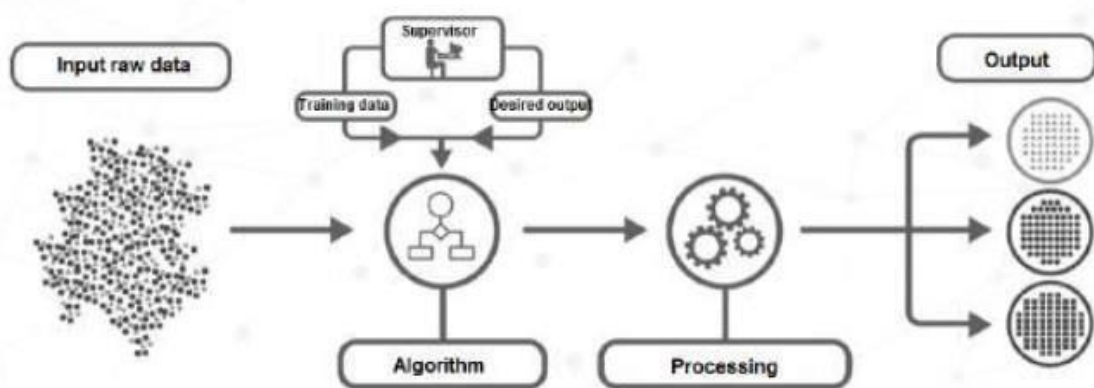


Figure 1.4: Supervised learning

Remarks

A “supervised learning” is so called because the process of an algorithm learning from the training dataset can be thought of as a teacher supervising the learning process. We know the correct answers (that is, the correct outputs), the algorithm iteratively makes predictions on the training data and is corrected by the teacher.

Learning stops when the algorithm achieves an acceptable level of performance.

Example

Consider the following data regarding patients entering a clinic. The data consists of the gender and age of the patients and each patient is labeled as “healthy” or “sick”.

gender	age	label
M	48	sick
M	67	sick
F	53	healthy
M	49	healthy
F	34	sick
M	21	healthy

Broadly, there are two types commonly used as supervised learning algorithms.

- 1) **Regression:** The output to be predicted is a continuous number in relevance with a given input dataset. Example use cases are predictions of retail sales, prediction of number of staff required for each shift, number of car parking spaces required for a retail store, credit score, for a customer, etc.
- 2) **Classification:** The output to be predicted is the actual or the probability of an event/class and the number of classes to be predicted can be two or more. The algorithm should learn the patterns in the relevant input of each class from historical data and be able to predict the unseen class or event in the future considering their input. An example use case is spam email filtering where the output expected is to classify an email into either a “spam” or “not spam.”

Building supervised learning machine learning models has three stages:

1. **Training:** The algorithm will be provided with historical input data with the mapped output. The algorithm will learn the patterns within the input data for each output

and represent that as a statistical equation, which is also commonly known as a model.

2. **Testing or validation:** In this phase the performance of the trained model is evaluated, usually by applying it on a dataset (that was not used as part of the training) to predict the class or event.
3. **Prediction:** Here we apply the trained model to a data set that was not part of either the training or testing. The prediction will be used to drive business decisions.

5. Write a note on Unsupervised learning.

There are situations where the desired output class/event is unknown for historical data. The objective in such cases would be to study the patterns in the input dataset to get better understanding and identify similar patterns that can be grouped into specific classes or events. As these types of algorithms do not require any intervention from the subject matter experts beforehand, they are called unsupervised learning.

Unsupervised learning is a type of machine learning algorithm used to draw inferences from datasets consisting of input data without labeled responses. In unsupervised learning algorithms, a classification or categorization is not included in the observations. There are no output values and so there is no estimation of functions. Since the examples given to the learner are unlabeled, the accuracy of the structure that is output by the algorithm cannot be evaluated. The most common unsupervised learning method is cluster analysis, which is used for exploratory data analysis to find hidden patterns or grouping in data.

Example

Consider the following data regarding patients entering a clinic.
The data consists of the gender and age of the patients.

gender	age
M	48
M	67
F	53
M	49
F	34
M	21

Based on this data, can we infer anything regarding the patients entering the clinic?

some examples of unsupervised learning.

- **Clustering:** Assume that the classes are not known beforehand for a given dataset. The goal here is to divide the input dataset into logical groups of related items. Some examples are grouping similar news articles, grouping similar customers based on their profile, etc.
- **Dimension Reduction:** Here the goal is to simplify a large input dataset by mapping them to a lower dimensional space. For example, carrying analysis on a large dimension dataset is very computational intensive, so to simplify you may want to find the key variables that hold a significant percentage (say 95%) of information and only use them for analysis.
- **Anomaly Detection:** Anomaly detection is also commonly known as outlier detection is the identification of items, events or observations which do not conform to an expected pattern or behavior in comparison with other items in a given dataset. It has applicability in a variety of domains,

such as machine or system health monitoring, event detection, fraud/intrusion detection etc.

6. Write a note on the following:

a. Semi-supervised learning

b. Reinforcement learning

Semi-supervised machine learning

This algorithms fall somewhere in between supervised and unsupervised learning, since they use both labeled and unlabeled data for training – typically a small amount of labeled data and a large amount of unlabeled data. The systems that use this method are able to considerably improve learning accuracy. Usually, semi-supervised learning is chosen when the acquired labeled data requires skilled and relevant resources in order to train it / learn from it. Otherwise, acquiring unlabeled data generally doesn't require additional resources.

Reinforcement Learning

The basic objective of reinforcement learning algorithms is to map situations to actions that yield the maximum final reward. While mapping the action, the algorithm should not just consider the immediate reward but also next and all subsequent rewards. For example, a program to play a game or drive a car will have to constantly interact with a dynamic environment in which it is expected to perform a certain goal.

This is somewhere between supervised and unsupervised learning. The algorithm gets told when the answer is wrong, but does not get told how to correct it. It has to explore and try out different possibilities until it works out how to get the answer right. Reinforcement learning is sometime called learning with a critic

because of this monitor that scores the answer, but does not suggest improvements.

Reinforcement learning is the problem of getting an agent to act in the world so as to maximize its rewards. A learner (the program) is not told what actions to take as in most forms of machine learning, but instead must discover which actions yield the most reward by trying them. In the most interesting and challenging cases, actions may affect not only the immediate reward but also the next situations and, through that, all subsequent rewards.

Examples of reinforcement learning techniques are the following:

- Markov decision process
- Q-learning
- Temporal Difference methods
- Monte-Carlo methods

7. Briefly explain Decision trees in machine learning.

Decision tree learning is a method for approximating discrete-valued target functions, in which the learned function is represented by a decision tree. Learned trees can also be re-represented as sets of if-then rules to improve human readability. These learning methods are among the most popular of inductive inference algorithms and have been successfully applied to a broad range of tasks from learning to diagnose medical cases to learning to assess credit risk of loan applicants.

Decision Tree Representation Decision trees classify instances by sorting them down the tree from the root to some leaf node, which provides the classification of the instance. Each node in the tree specifies a test of some attribute of the instance, and each

branch descending from that node corresponds to one of the possible values for this attribute. An instance is classified by starting at the root node of the tree, testing the attribute specified by this node, then moving down the tree branch corresponding to the value of the attribute in the given example. This process is then repeated for the subtree rooted at the new node

Here the target attribute PlayTennis, which can have values yes or no for different Saturday mornings, is to be predicted based on other attributes of the morning in question

Day	<i>Outlook</i>	<i>Temperature</i>	<i>Humidity</i>	<i>Wind</i>	<i>PlayTennis</i>
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

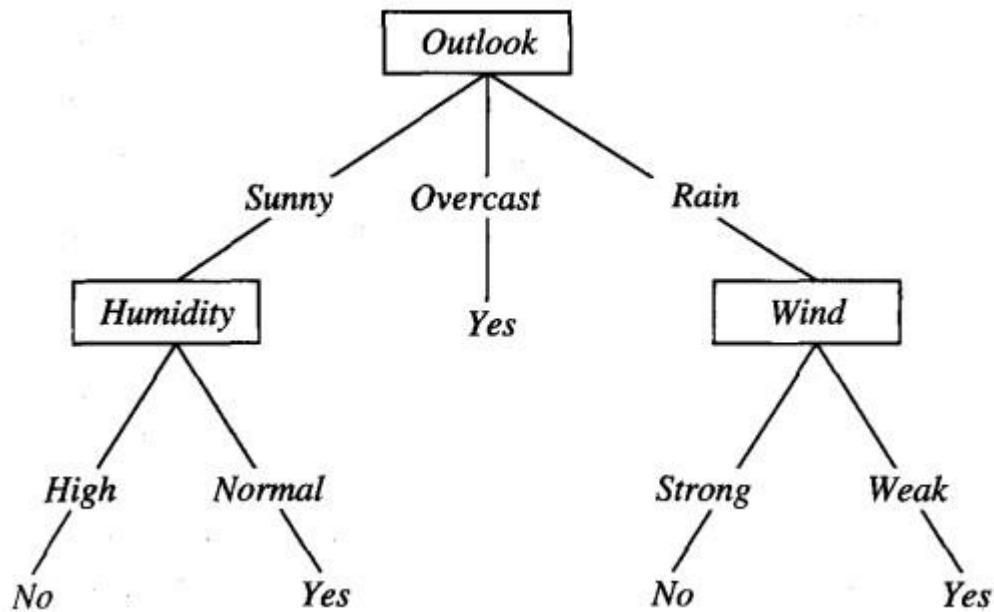


FIGURE above shows A decision tree for the concept Play Tennis. An example is classified by sorting it through the tree to the appropriate leaf node, then returning the classification associated with this leaf (in this case, Yes or No). This tree classifies Saturday mornings according to whether or not they are suitable for playing tennis.

8. Write the steps in ID3 algorithm to construct the decision trees.

ID3

ID3 Algorithm will perform following tasks recursively

1. Create root node for the tree
2. If all examples are positive, return leaf node „positive“
3. Else if all examples are negative, return leaf node „negative“
4. Calculate the entropy of current state $H(S)$
5. For each attribute, calculate the entropy with respect to the attribute „x“ denoted by $H(S, x)$
6. Select the attribute which has maximum value of $IG(S, x)$
7. Remove the attribute that offers highest IG from the set of attributes

8. Repeat until we run out of all attributes, or the decision tree has all leaf nodes.

9. List and explain the terminologies related to regression analysis

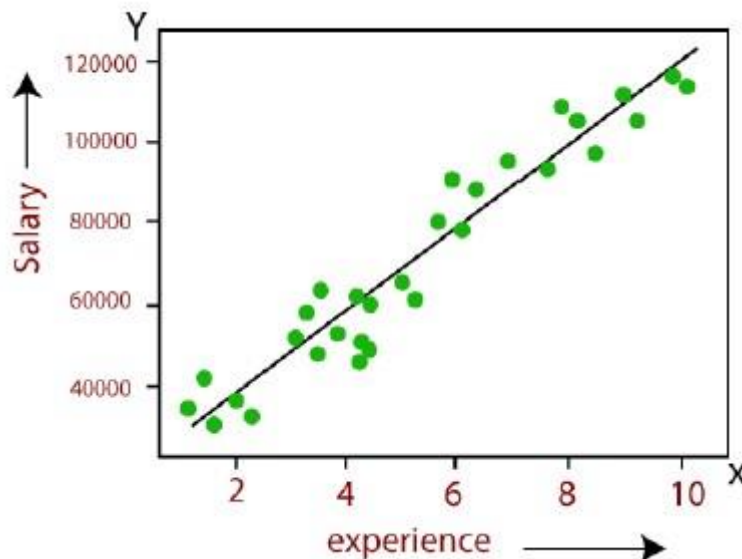
Terminologies Related to the Regression Analysis:

- **Dependent Variable:** The main factor in Regression analysis which we want to predict or understand is called the dependent variable. It is also called **target variable**.
- **Independent Variable:** The factors which affect the dependent variables or which are used to predict the values of the dependent variables are called independent variable, also called as a **predictor**.
- **Outliers:** Outlier is an observation which contains either very low value or very high value in comparison to other observed values. An outlier may hamper the result, so it should be avoided.
- **Multicollinearity:** If the independent variables are highly correlated with each other than other variables, then such condition is called Multicollinearity. It should not be present in the dataset, because it creates problem while ranking the most affecting variable.
- **Underfitting and Overfitting:** If our algorithm works well with the training dataset but not well with test dataset, then such problem is called **Overfitting**. And if our algorithm does not perform well even with training dataset, then such problem is called **underfitting**.

10. Briefly explain Linear Regression.

- Linear regression is a statistical regression method which is used for predictive analysis.

- It is one of the very simple and easy algorithms which works on regression and shows the relationship between the continuous variables.
- It is used for solving the regression problem in machine learning.
- Linear regression shows the linear relationship between the independent variable (X-axis) and the dependent variable (Y-axis), hence called linear regression.
- If there is only one input variable (x), then such linear regression is called **simple linear regression**. And if there is more than one input variable, then such linear regression is called **multiple linear regression**.
- The relationship between variables in the linear regression model can be explained using the below image. Here we are predicting the salary of an employee on the basis of **the year of experience**.



Below is the mathematical equation for Linear regression:

$$Y = aX + b$$

Here,

Y = dependent variables (target variables), X = Independent

variables (predictor variables), a and b are the linear coefficients

Some popular applications of linear regression are:

- Analyzing trends and sales estimates
- Salary forecasting
- Real estate prediction
- Arriving at ETAs in traffic.

In non-linear regression, the data can't be modelled to fit a straight line i.e. modelled to fit a curve as shown in below figure.

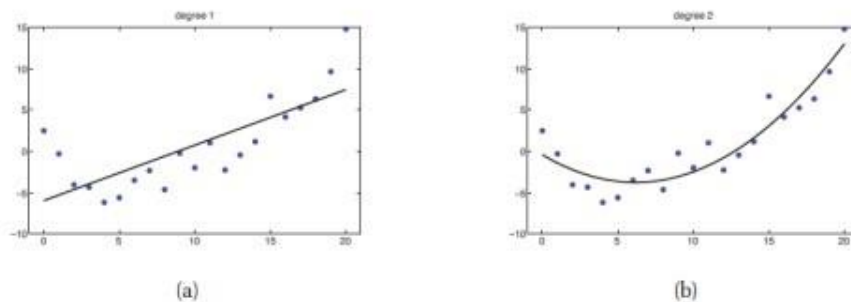


Figure (a) Linear regression on some 1d data. (b) Same data with polynomial regression (degree 2).

For example the equation can have the form: X^{Theta}

Here are some examples of real-world regression problems.

- Predict tomorrow's stock market price given current market conditions and other possible side information.
- Predict the age of a viewer watching a given video on YouTube. .

11. Briefly explain Logistic Regression.

- Logistic regression is another supervised learning algorithm which is used to solve the classification problems. In **classification problems**, we have dependent variables in a binary or discrete format such as 0 or 1.

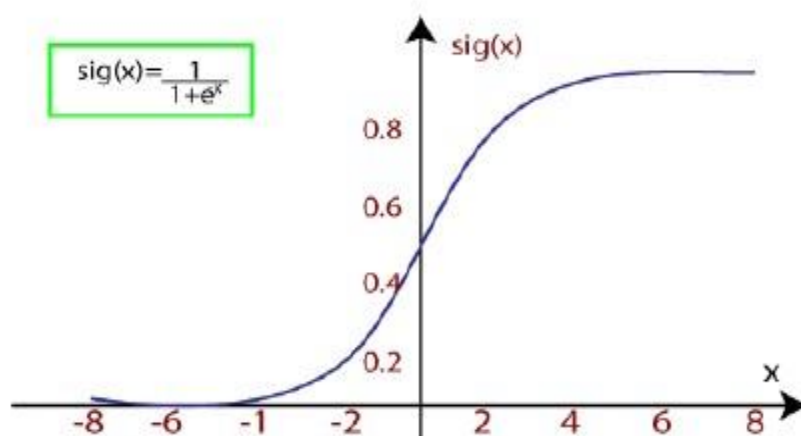
- Logistic regression algorithm works with the categorical variable such as 0 or 1, Yes or No, True or False, Spam or not spam, etc.
- It is a predictive analysis algorithm which works on the concept of probability.
- Logistic regression is a type of regression, but it is different from the linear regression algorithm in the term how they are used.
- Logistic regression uses **sigmoid function** or logistic function which is a complex cost function. This sigmoid function is used to model the data in logistic regression. The function can be represented as:

$$f(x) = \frac{1}{1+e^{-x}}$$

$f(x)$ = Output between the 0 and 1 value.

x = input to the
function e = base
of natural
logarithm.

When we provide the input values (data) to the function, it gives the S-curve as follows:



It uses the concept of threshold levels, values above the threshold level are rounded up to 1, and values below the threshold level are rounded up to 0.

There are three types of logistic regression:

- Binary(0/1, pass/fail)
- Multi(cats, dogs, lions)
- Ordinal(low, medium, high)

12. Write a note on Artificial Neural Network.

Artificial neural networks (ANNs) provide a general, practical method for learning real-valued, discrete-valued, and vector-valued functions from examples. For certain types of problems, such as learning to interpret complex real-world sensor data, artificial neural networks are among the most effective learning methods currently known.

An artificial neural network (ANN) is like a computing system that simulates how the human brain analyzes information and processes it. It is the branch of artificial intelligence (AI) and solves problems that may be difficult or impossible to solve such issues for humans. In addition, ANNs have the potential for self-learning that provide better results if more data becomes available.

Artificial neurons consist of the following things:

Interconnecting model of neurons. The neuron is the elementary component connected with other neurons to form the network.

Learning algorithm to train the network. Various learning algorithms are available in the literature to train the model. Each layer consists of neurons, and these neurons are connected to other layers. Weight is also assigned to each layer, and these weights are changed at each iteration for training purpose.

An artificial neural network (ANN) consists of an interconnected group of artificial neurons that process information through input, hidden and output layers and use a connectionist approach to computation. Neural networks use nonlinear statistical data modeling tools to solve complex problems by finding the complex relationships between inputs and outputs. After getting this relation, we can predict the outcome or classify our problems.

Figure 2 describes three neurons that perform "AND" logical operations. In this case, the output neuron will fire if both input neurons are fired. The output neurons use a threshold value (T), $T=3/2$ in this case. If none or only one input neuron is fired, then the total input to the output becomes less than 1.5 and firing for output is not possible. Take another scenario where both input neurons are firing, and the total input becomes $1+1=2$, which is greater than the threshold value of 1.5, then output neurons will fire. Similarly, we can perform the "OR" logical operation with the help of the same architecture but set the new threshold to 0.5. In this case, the output neurons will be fired if at least one input is fired.

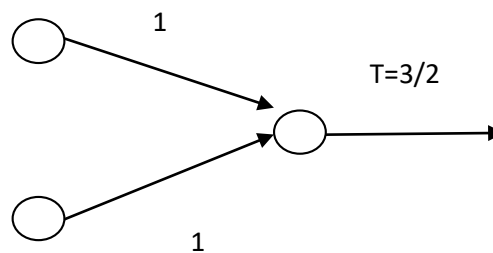
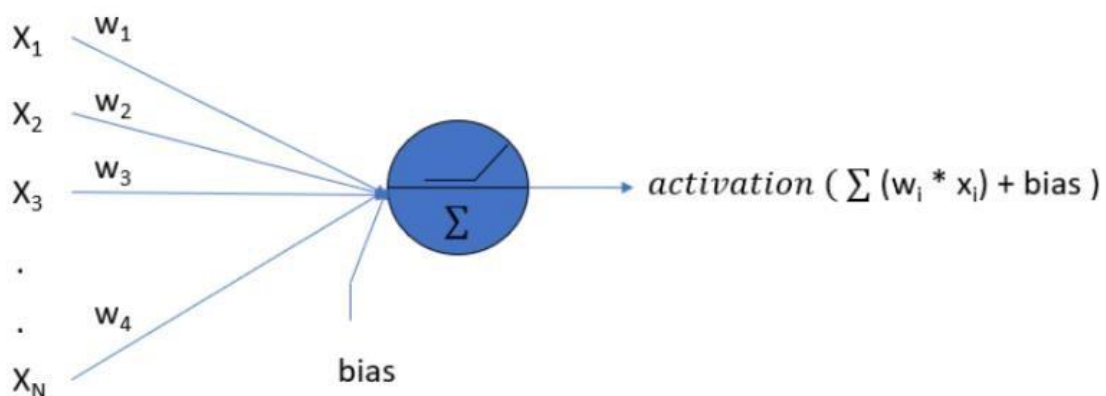


Figure 2:Three Neurons Diagram



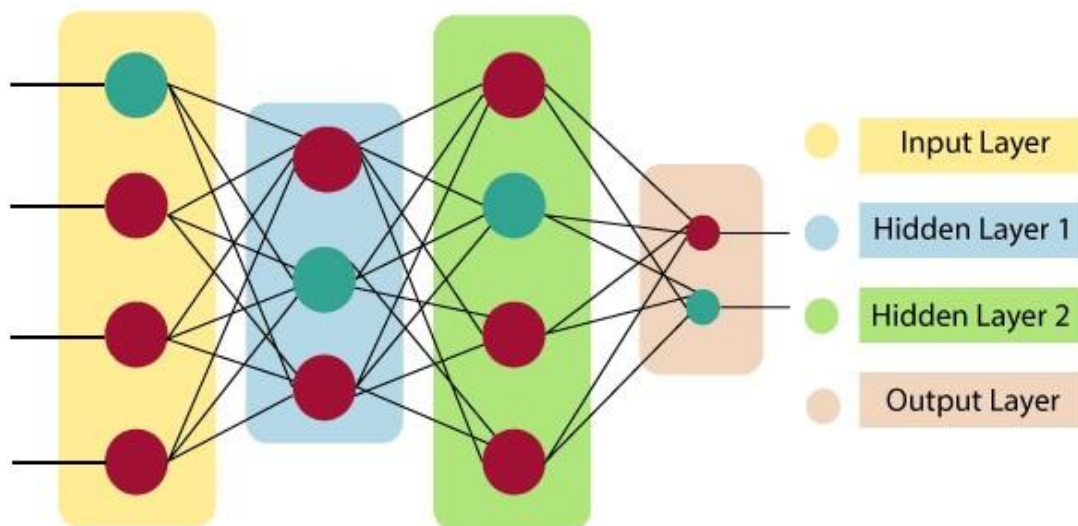
A single neuron shown with X_i inputs with their respective weights W_i and a bias term and applied activation function

Artificial Neural Network has a huge number of interconnected processing elements, also known as Nodes. These nodes are connected with other nodes using a connection link. The connection link contains weights, these weights contain the information about the input signal. Each iteration and input in turn leads to updation of these weights. After inputting all the data instances from the training data set, the final weights of the Neural Network along with its architecture is known as the Trained Neural Network. This process is called Training of Neural Networks. These trained neural networks solve specific problems as defined in the problem statement.

Types of tasks that can be solved using an artificial neural network include Classification problems, Pattern Matching, Data Clustering, etc.

13.Explain the architecture of an Artificial Neural Network.

The architecture of an artificial neural network:



Input Layer: As the name suggests, it accepts inputs in several different formats provided by the programmer.

Hidden Layer: The hidden layer presents in-between input and output layers. It performs all the calculations to find hidden features and patterns.

Output Layer: The input goes through a series of transformations using the hidden layer, which finally results in output that is conveyed using this layer. The artificial neural network takes input and computes the weighted sum of the inputs and includes a bias. This computation is represented in the form of a transfer function.

$$\sum_{i=1}^n W_i * X_i + b$$

It determines weighted total is passed as an input to an activation function to produce the output. Activation functions choose whether a node should fire or not. Only those who are fired make it to the output layer. There are distinctive activation functions available that can be applied upon the sort of task we are performing.

14. Briefly explain the advantages and disadvantages of Artificial Neural Network.

Advantages of Artificial Neural Network (ANN)

- **Parallel processing capability:** Artificial neural networks have a numerical value that can perform more than one task simultaneously.
- **Storing data on the entire network:** Data that is used in traditional programming is stored on the whole network, not on a database. The disappearance of a couple of pieces of data in one place doesn't prevent the network from working.
- **Capability to work with incomplete knowledge:** After ANN training, the information may produce output even with

inadequate data. The loss of performance here relies upon the significance of missing data.

- **Having a memory distribution:** For ANN is to be able to adapt, it is important to determine the examples and to encourage the network according to the desired output by demonstrating these examples to the network. The succession of the network is directly proportional to the chosen instances, and if the event can't appear to the network in all its aspects, it can produce false output.
- **Having fault tolerance:** Extortion of one or more cells of ANN does not prohibit it from generating output, and this feature makes the network fault tolerance.

Disadvantages of Artificial Neural Network:

- **Assurance of proper network structure:** There is no particular guideline for determining the structure of artificial neural networks. The appropriate network structure is accomplished through experience, trial, and error.
- **Unrecognized behavior of the network:** It is the most significant issue of ANN. When ANN produces a testing solution, it does not provide insight concerning why and how. It decreases trust in the network.
- **Hardware dependence:** Artificial neural networks need processors with parallel processing power, as per their structure. Therefore, the realization of the equipment is dependent.
- **Difficulty of showing the issue to the network:** ANNs can work with numerical data. Problems must be converted into numerical values before being introduced to ANN. The presentation mechanism to be resolved here will directly impact the performance of the network. It relies on the user's abilities.

- **The duration of the network is unknown:** The network is reduced to a specific value of the error, and this value does not give us optimum results. *“Science artificial neural networks that have steeped into the world in the mid-20th century are exponentially developing. In the present time, we have investigated the pros of artificial neural networks and the issues encountered in the course of their utilization. It should not be overlooked that the cons of ANN networks, which area flourishing science branch, are eliminated individually, and their pros are increasing day by day. It means thatartificial neural networks will turn into an irreplaceable part of our lives progressively important.”*

15.List and explain the types of Artificial Neural Network.

There are various types of Artificial Neural Networks (ANN) depending upon the human brain neuron and network functions, an artificial neural network similarly performs tasks. The majority of the artificial neural networks will have some similarities with a more complex biological partner and are very effective at their expected tasks. For example, segmentation or classification.

- **Feedback ANN:** In this type of ANN, the output returns into the network to accomplish the best-evolved results internally. As per the **University of Massachusetts**, Lowell Centre for Atmospheric Research. The feedback networks feed information back into itself and are well suited to solve optimization issues. The Internal system error corrections utilize feedback ANNs.
- **Feed-Forward ANN:** A feed-forward network is a basic neural network comprising of an input layer, an output layer, and at least one layer of a neuron. Through assessment of its output by reviewing its input, the intensity of the network can be noticed based on group behavior of the

associated neurons, and the output is decided. The primary advantage of this network is that it figures out how to evaluate and recognize input patterns.

- **Prerequisite**

No specific expertise is needed as a prerequisite before starting this tutorial.

- **Audience**

Our Artificial Neural Network Tutorial is developed for beginners as well as professionals, to help them understand the basic concept of ANNs.

16. Write a note on Autoencoders.

Autoencoders: As the name suggests, an autoencoder aims to learn encoding as a representation of training sample data automatically without human intervention. The autoencoder is widely used for dimensionality reduction and data de-noising.

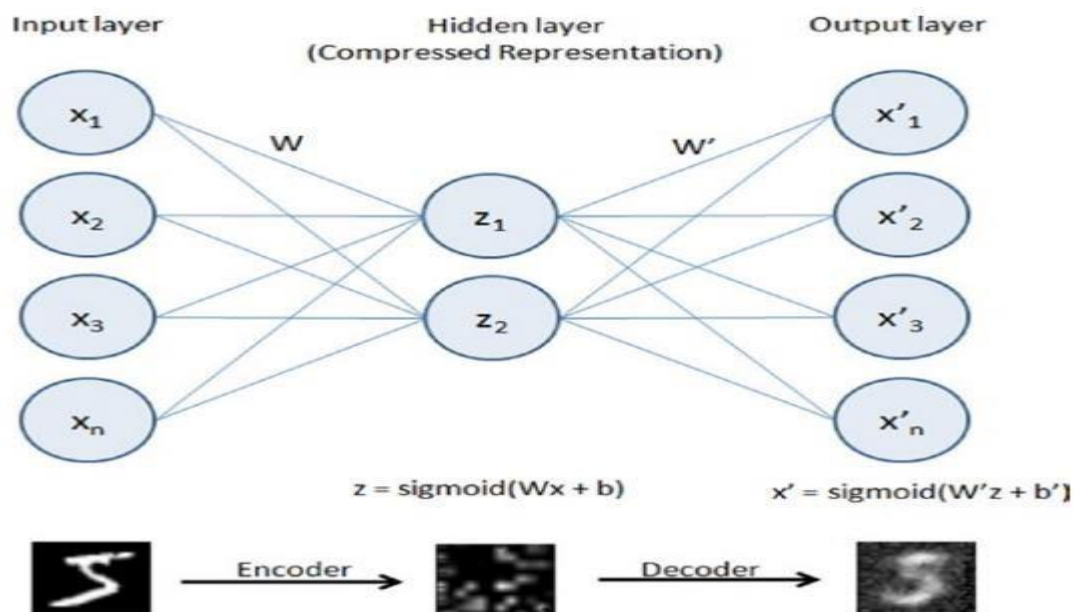


Figure 6-5. Autoencoder

Building an autoencoder will typically have three elements.

- Encoding function to map input to a hidden representation through a nonlinear function, $z = \text{sigmoid}(Wx + b)$.

- A decoding function such as $x' = \text{sigmoid}(W'y + b')$, which will map back into reconstruction x' with same shape as x .
- A loss function, which is a distance function to measure the information loss between the compressed representation of data and the decompressed representation. Reconstruction error can be measured using traditional squared error $\|x - z\|^2$.

17. Write a note on the following:

- CNN**
- RNN**
- LSTM**

Convolution Neural Network (CNN)

CNN's are similar to ordinary neural networks, except that it explicitly assumes that the inputs are images, which allows us to encode certain properties into the architecture. These then make the forward function efficient to implement and reduces the parameters in the network. The neurons are arranged in three dimensions: width, height, and depth

Recurrent Neural Network (RNN)

The MLP (feedforward network) is not known to do well on sequential events models such as the probabilistic language model of predicting the next word based on the previous word at every given point. RNN architecture addresses this issue. It is similar to MLP except that they have a feedback loop, which means they feed previous time steps into the current step. This type of architecture generates sequences to simulate situation and create synthetic data, making them the ideal modeling choice to work on sequence data such as speech text mining, image captioning, time series prediction, robot control, language modeling, etc.

Long Short-Term Memory (LSTM)

LSTM is an implementation of improved RNN architecture to address the issues of general RNN, and it enables long-range dependencies. It is designed to have better memory through linear memory cells surrounded by a set of gate units used to control the flow of information, when information should enter the memory, when to forget, and when to output. It uses no activation function within its recurrent components, thus the gradient term does not vanish with back propagation.

LSTM components are:

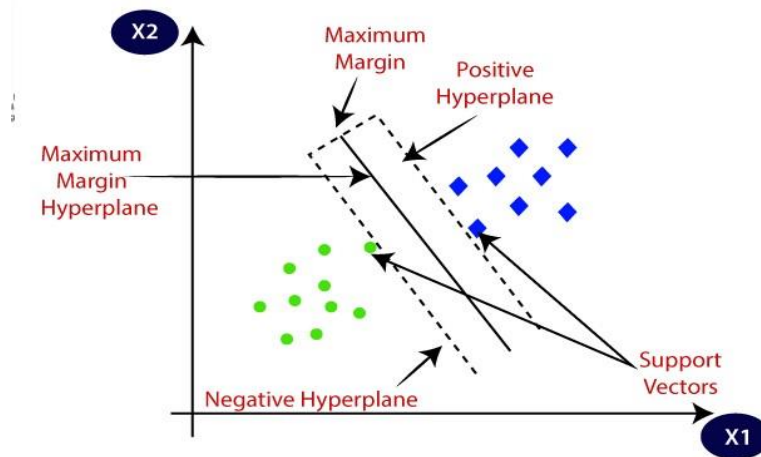
- **Input gate layer:** This decides which values to store in the cell state
- **Forget gate layer:** As the name suggested this decides what information to throw away from the cell state
- **Output gate layer:** Create a vector of values that can be added to the cell state.
- **Memory cell state vector**

18. Write a note on Support Vector Machine.

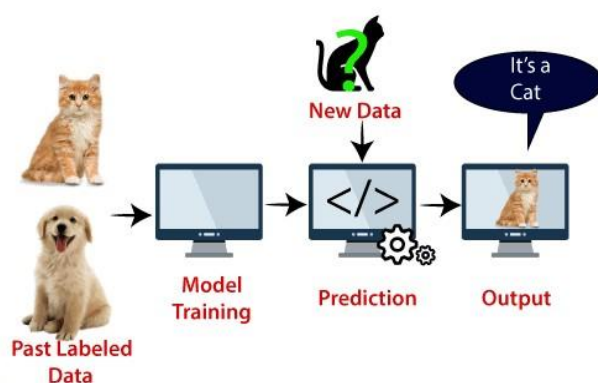
Support Vector Machine or SVM is one of the most popular Supervised Learning algorithms, which is used for Classification as well as Regression problems. However, primarily, it is used for Classification problems in Machine Learning.

The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in the future. This best decision boundary is called a hyperplane.

SVM chooses the extreme points/vectors that help in creating the hyperplane. These extreme cases are called as support vectors, and hence algorithm is termed as Support Vector Machine. Consider the below diagram in which there are two different categories that are classified using a decision boundary or hyperplane:



Example: SVM can be understood with the example that we have used in the KNN classifier. Suppose we see a strange cat that also has some features of dogs, so if we want a model that can accurately identify whether it is a cat or dog, so such a model can be created by using the SVM algorithm. We will first train our model with lots of images of cats and dogs so that it can learn about different features of cats and dogs, and then we test it with this strange creature. So as support vector creates a decision boundary between these two data (cat and dog) and choose extreme cases (support vectors), it will see the extreme case of cat and dog. On the basis of the support vectors, it will classify it as a cat. Consider the below diagram:



SVM algorithm can be used for **Face detection, image classification, text categorization**, etc.

19. Differentiate Supervised learning and Unsupervised learning.

Supervised Learning	Unsupervised Learning
Supervised learning algorithms are trained using labeled data.	Unsupervised learning algorithms are trained using unlabeled data.
Supervised learning model takes direct feedback to check if it is predicting correct output or not.	Unsupervised learning model does not take any feedback.
Supervised learning model predicts the output.	Unsupervised learning model finds the hidden patterns in data.
In supervised learning, input data is provided to the model along with the output.	In unsupervised learning, only input data is provided to the model.
The goal of supervised learning is to train the model so that it can predict the output when it is given new data.	The goal of unsupervised learning is to find the hidden patterns and useful insights from the unknown dataset.
Supervised learning needs supervision to train the model.	Unsupervised learning does not need any supervision to train the model.
Supervised learning can be categorized in Classification and Regression problems.	Unsupervised Learning can be classified in Clustering and Associations problems.

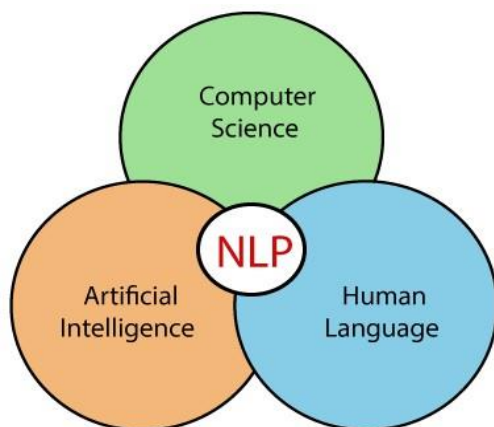
Supervised learning can be used for those cases where we know the input as well as corresponding outputs.	Unsupervised learning can be used for those cases where we have only input data and no corresponding output data.
Supervised learning model produces an accurate result.	Unsupervised learning model may give less accurate result as compared to supervised learning.
Supervised learning is not close to true Artificial intelligence as in this, we first train the model for each data, and then only it can predict the correct output.	Unsupervised learning is more close to the true Artificial Intelligence as it learns similarly as a child learns daily routine things by his experiences.
It includes various algorithms such as Linear Regression, Logistic Regression, Support Vector Machine, Multi-class Classification, Decision tree, Bayesian Logic, etc.	It includes various algorithms such as Clustering, KNN, and Apriori algorithm.

20. List and explain the applications of artificial Intelligence.

(NLP, Text Classification, Information Retrieval, Speech recognition, Image Processing And Computer Vision, Robotics)

21. Write a note on NLP.

NLP stands for **Natural Language Processing**, which is a part of **Computer Science**, **Human language**, and **Artificial Intelligence**. It is the technology that is used by machines to understand, analyse, manipulate, and interpret human's languages. It helps developers to organize knowledge for performing tasks such as **translation, automatic summarization, Named Entity Recognition (NER), speech recognition, relationship extraction, and topic segmentation**.



Components of NLP

There are the following two components of NLP –

- 1. Natural Language Understanding (NLU):** Natural Language Understanding (NLU) helps the machine to understand and analyse human language by extracting the metadata from content such as concepts, entities, keywords, emotion, relations, and semantic roles.

NLU mainly used in Business applications to understand the customer's problem in both spoken and written language.

NLU involves the following tasks -
o It is used to map the given input into useful representation.

o It is used to analyze different aspects of the language.

- 2. Natural Language Generation (NLG):** Natural Language Generation (NLG) acts as a translator that converts the

computerized data into natural language representation. It mainly involves Text planning, Sentence planning, and Text Realization.

22.Explain the phases of NLP.

There are the following five phases of NLP:

1. Lexical Analysis and Morphological: The first phase of NLP is the Lexical Analysis. This phase scans the source code as a stream of characters and converts it into meaningful lexemes. It divides the whole text into paragraphs, sentences, and words.

2. Syntactic Analysis (Parsing): Syntactic Analysis is used to check grammar, word arrangements, and shows the relationship among the words.

Example: Agra goes to the Poonam

In the real world, Agra goes to the Poonam, does not make any sense, so this sentence is rejected by the Syntactic analyzer.

3. Semantic Analysis: Semantic analysis is concerned with the meaning representation. It mainly focuses on the literal meaning of words, phrases, and sentences.

4. Discourse Integration: Discourse Integration depends upon the sentences that proceeds it and also invokes the meaning of the sentences that follow it.

5. Pragmatic Analysis: Pragmatic is the fifth and last phase of NLP. It helps you to discover the intended effect by applying a set of rules that characterize cooperative dialogues.

For Example: "Open the door" is interpreted as a request instead of an order.

23.Explain the applications of NLP.

- Question Answering: Question Answering focuses on building systems that automatically answer the questions asked by humans in a natural language.
- Spam Detection: Spam detection is used to detect unwanted e-mails getting to a user's inbox.
- Sentiment Analysis: Sentiment Analysis is also known as opinion mining. It is used on the web to analyse the attitude, behaviour, and emotional state of the sender. This application is implemented through a combination of NLP (Natural Language Processing) and statistics by assigning the values to the text (positive, negative, or natural), identify the mood of the context (happy, sad, angry, etc.)
- Machine Translation: Machine translation is used to translate text or speech from one natural language to another natural language.Example: Google Translator
- Spelling correction: Microsoft Corporation provides word processor software like MS-word, PowerPoint for the spelling correction.
- Speech Recognition: Speech recognition is used for converting spoken words into text. It is used in applications, such as mobile, home automation, video recovery, dictating to Microsoft Word, voice biometrics, voice user interface, and so on.
- Chatbot: Implementing the Chatbot is one of the important applications of NLP. It is used by many companies to provide the customer's chat services.
- Information extraction: Information extraction is one of the most important applications of NLP. It is used for extracting structured information from unstructured or semi-structured machine-readable documents.
- Natural Language Understanding (NLU): It converts a large set of text into more formal representations such as first-order logic structures that are easier for the computer programs to manipulate notations of the natural language processing.

24.Briefly explain the steps involved in Text Classification process.

Step1: Data collection

Collect a set of text documents with their corresponding categories for the text labeling process. Gathering data is the most important step in solving any supervised machine learning problem. Your text classifier can only be as good as the dataset it is built from.

Here are some important things to remember when collecting data:

- The more training examples (referred to as *samples* in the rest of this guide) you have, the better. This will help your model generalize better.
- Make sure the number of samples for every *class* or topic is not overly *imbalanced*. That is, you should have comparable number of samples in each class.
- Make sure that your samples adequately cover the *space of possible inputs*, not only the common cases.

Step 2: Data preprocessing

Clean and prepare the text data by removing unnecessary symbols, converting to lowercase, and handling special characters such as punctuation. Understanding the characteristics of your data beforehand will enable you to build a better model. This could simply mean obtaining a higher accuracy. It's good practice to run some checks on it: pick a few samples and manually check if they are consistent with your expectations.

Step3:Tokenization

Break the text apart into tokens, which are small units like words. Tokens help find matches and connections by creating individually searchable parts. This step is especially useful for **vector search** and **semantic search**, which give results based on user intent.

Step 4: Feature extraction

Convert the text into numerical representations that machine learning models can understand. Some common methods include counting the occurrences of words (also known as Bag-of-Words) or using **word embeddings** to capture word meanings. When we convert all of the texts in a dataset into tokens, we may end up with tens of thousands of tokens. Not all of these tokens/features contribute to label prediction. So we can drop certain tokens, for instance those that occur extremely rarely across the dataset. We can also measure feature importance (how much each token contributes to label predictions), and only include the most informative tokens.

Step5:Build, Train and evaluate model:

Now that the data is clean and preprocessed, you can use it to train a machine learning model. The model will learn patterns and associations between the text's features and their categories. This helps it understand the text labeling conventions using the pre-labeled examples. Training involves making a prediction based on the current state of the model, calculating how incorrect the prediction is, and updating the weights or parameters of the network to minimize this error and make the model predict better. We repeat this process until our model has converged and can no longer learn. There are three key parameters to be chosen for this process

- **Metric:** How to measure the performance of our model using a ***metric***. We used *accuracy* as the metric in our experiments.
- **Loss function:** A function that is used to calculate a loss value that the training process then attempts to minimize by tuning the network weights. For classification problems, cross-entropy loss works well.
- **Optimizer:** A function that decides how the network weights will be updated based on the output of the loss function. We used the popular Adam optimizer in our experiments

Step 6: Text labeling

Create a new, separate dataset to start text labeling and classifying new text. In the text labeling process, the model separates the text into the predetermined categories from the data collection step.

Step 8: Hyperparameter tuning

Depending on how the model evaluation goes, you may want to adjust the model's settings to optimize its performance. We had to choose a number of hyperparameters for defining and training the model. We relied on intuition, examples and best practice recommendations. Our first choice of hyperparameter values, however, may not yield the best results. It only gives us a good starting point for training. Every problem is different and tuning these hyperparameters will help refine our model to better represent the particularities of the problem at hand.

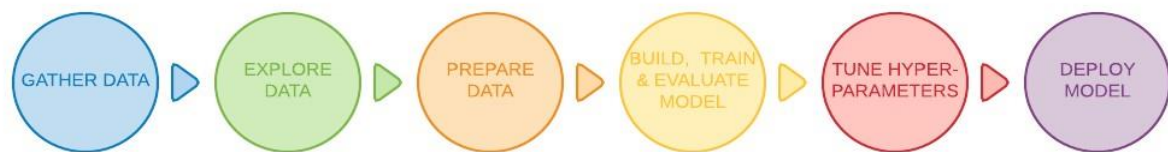
some of the hyperparameters :

- **Number of layers in the model:** The number of layers in a neural network is an indicator of its complexity.

- **Number of units per layer:** The units in a layer must hold the information for the transformation that a layer performs.
- **Dropout rate:** Dropout layers are used in the model

Step 9: Model deployment

Use the trained and tuned model to classify new text data into their appropriate categories.



Workflow for solving machine learning problems

25. Briefly explain the Information retrieval Model with its types.

A retrieval model (IR) chooses and ranks relevant pages based on a user's query.

Document selection and ranking can be formalized using matching functions

that return retrieval status values (RSVs) for each document in a collection since documents and queries are written in the same way. The majority of IR systems portray document contents using a collection of descriptors known as words from a vocabulary V .

The query-document matching function in an IR model is defined in the following ways:

- The estimation of the likelihood of user relevance for each page and query in relation to a collection of q training documents.
- In a vector space, the similarity function between queries and documents is computed.

Types Of Information Retrieval Models

1. Classic IR Model

It is the most basic and straightforward IR model. This paradigm is founded on mathematical information that was easily recognized and comprehended. The three traditional IR models are Boolean, Vector, and Probabilistic.

2. Non-Classic IR Model

It is diametrically opposed to the traditional IR model. Addition than probability, similarity, and Boolean operations, such IR models are based on other ideas. Non-classical IR models include situation theory models, information logic models, and interaction models.

3. Alternative IR Model

It is an improvement to the traditional IR model that makes use of some unique approaches from other domains. Alternative IR models include fuzzy models, cluster models, and latent semantic indexing (LSI) models

26. Explain the working of Speech Recognition Technology.

Speech recognition is a software technology fueled by cutting-edge solutions like Natural Language Processing (NLP) and Machine Learning (ML). NLP, an AI system that analyses natural human speech, is sometimes referred to as human language processing. The vocal data is first transformed into a digital format that can be processed by computer software. Then, the digitized data is subjected to additional processing using NLP, ML, and deep learning techniques. Consumer products like smartphones, smart homes, and other voiceactivated solutions can employ this digitized speech.

A piece of software operates four procedures to convert the audio that a microphone records into text that both computers and people can understand:

1. Analyze the audio;
2. Separate it into sections;

3. Create a computer-readable version of it using digitization, and
4. The most appropriate text representation should be found using an algorithm.

Due to how context-specific and extremely varied human speech is, voice recognition algorithms must adjust. Different speech patterns, speaking styles, languages, dialects, accents, and phrasings are used to train the software algorithms that process and organize audio into text. The software also distinguishes speech sounds from the frequently present background noise.

Speech recognition systems utilize one of two types of models to satisfy these requirements:

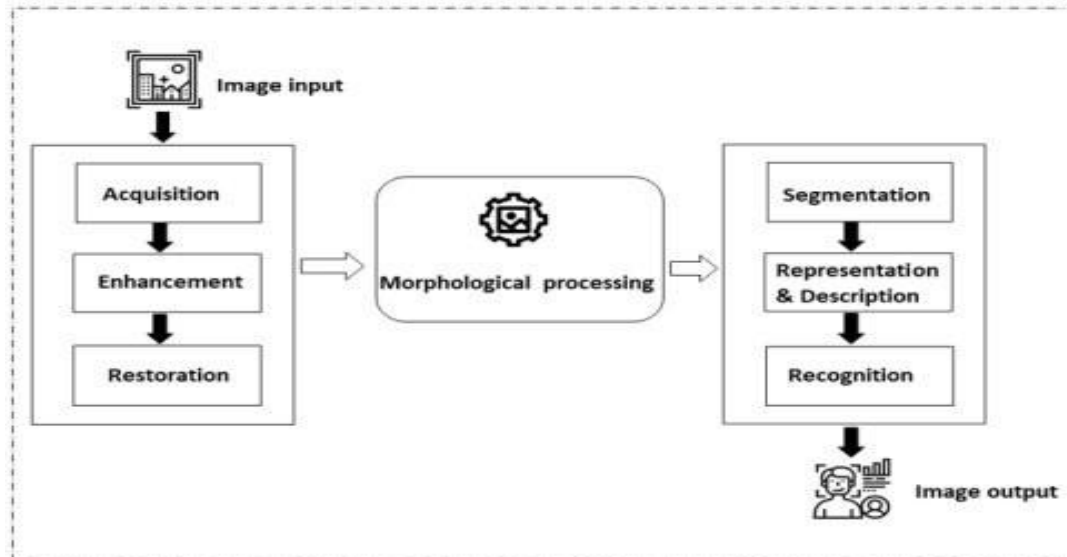
- Good modeling. These illustrate the connection between speech linguistics and audio signals.
- Language layouts. Here, word sequences and sounds are matched to identify words with similar sounds.

27.Explain the different speech recognition algorithms.

- **Hidden Markov model (HMM).** HMM is an algorithm that handles speech diversity, such as pronunciation, speed, and accent. It provides a simple and effective framework for modeling the temporal structure of audio and voice signals and the sequence of phonemes that make up a word. For this reason, most of today's speech recognition systems are based on an HMM.
- **Dynamic time warping (DTW).** DTW is used to compare two separate sequences of speech that are different in speed. For example, you have two audio recordings of someone saying "good morning" – one slow, one fast. In this case, the DTW algorithm can sync the two recordings, even though they differ in speed and length.
- **Artificial neural networks (ANN).** ANN is a computational model used in speech recognition applications that helps computers understand spoken human language. It uses deep learning techniques and basically imitates the patterns of how neural networks work in the human brain, which allows the computer to make decisions in a human-like manner.

28.Explain the elements of Image processing.

AI image processing works through a combination of advanced algorithms, neural networks, and data processing to analyze, interpret, and manipulate digital images. Here's a simplified overview of how AI image processing works:



Acquisition of image: The initial level begins with image pre-processing which uses a sensor to capture the image and transform it into a usable format.

Enhancement of image: Image enhancement is the technique of bringing out and emphasising specific interesting characteristics which are hidden in an image.

Restoration of image: Image restoration is the process of enhancing an image's look. Picture restoration, as opposed to image augmentation, is carried out utilising specific mathematical or probabilistic models.

Colour image processing: A variety of digital colour modelling approaches such as HSI (Hue-Saturation-Intensity), CMY (Cyan-Magenta-Yellow) and RGB (Red-Green-Blue) etc. are used in colour picture processing.

Compression and decompression of image: This enables adjustments to image resolution and size, whether for image reduction or restoration depending on the situation, without lowering image quality below a desirable level. Lossy and lossless compression techniques are the two main types of image file compression which are being employed in this stage.

Morphological processing: Digital images are processed depending on their shapes using an image processing technique known as morphological operations. The operations depend on the pixel values rather than their numerical values, and well suited for the processing of binary images. It aids in removing imperfections for structure of the image.

Segmentation, representation and description: The segmentation process divides a picture into segments, and each segment is represented and described in such a way that it can be processed further by a computer. The image's quality and regional characteristics are covered by representation. The description's job is to extract quantitative data that helps distinguish one class of items from another.

Recognition of image: A label is given to an object through recognition based on its description. Some of the often-employed algorithms in the process of recognising images include the Scale-invariant Feature Transform (SIFT), the Speeded Up Robust Features (SURF), and the PCA (Principal Component Analysis)

29. Write a note on Computer vision.

Computer vision is a field of artificial intelligence that enables computers to derive meaningful information from digital images, videos, and other visual inputs . It uses machine learning algorithms to analyze and interpret images, and it has a wide range of applications in various fields.

Computer vision works by using deep learning algorithms to analyze and interpret images. These algorithms are trained on large datasets of images and use convolutional neural networks (CNNs) to identify patterns and features in the images .Once the patterns and features are identified, the computer can use them to recognize objects, classify images, and perform other image-based tasks.

Some of the applications of computer vision include:

1. **Autonomous Vehicles:** Computer vision is used in autonomous vehicles to detect and recognize objects such as pedestrians, other vehicles, and traffic signals. This helps the vehicle navigate safely and avoid accidents
2. **Retail:** Computer vision is used in retail to track inventory, monitor customer behavior, and analyze shopping patterns.

It can also be used to personalize the shopping experience for customers

3. **Healthcare:** Computer vision is used in healthcare to diagnose diseases, monitor patients, and assist with surgical procedures. It can also be used to analyze medical images such as X-rays and MRIs
4. **Security:** Computer vision is used in security to detect and recognize faces, track individuals, and monitor public spaces. It can also be used to detect suspicious behavior and prevent crime .
5. **Entertainment:** Computer vision is used in entertainment to create special effects, enhance video games, and develop virtual reality experiences. It can also be used to analyze audience reactions and preferences.

30.Differentiate Image processing and Computer Vision.

Image Processing	Computer Vision
Image processing is mainly focused on processing the raw input images to enhance them or preparing them to do other tasks	Computer vision is focused on extracting information from the input images or videos to have a proper understanding of them to predict the visual input like human brain.
Image processing uses methods like Anisotropic diffusion, Hidden Markov models, Independent component analysis, Different Filtering etc.	Image processing is one of the methods that is used for computer vision along with other Machine learning techniques, CNN etc.
Image Processing is a subset of Computer Vision.	Computer Vision is a superset of Image Processing.
Examples of some Image Processing applications are- Rescaling image (Digital Zoom), Correcting illumination, Changing tones etc.	Examples of some Computer Vision applications are- Object detection, Face detection, Hand writing recognition etc.

31. Write a note on Robotics.

Robotics is a field of engineering and science that involves the design, construction, operation, and use of robots. Robots are autonomous or semiautonomous machines that can perform tasks in the physical world.

AI allows robots to interpret and respond to their environment. Advanced sensors combined with AI algorithms enable robots to perceive and understand the world around them using vision, touch, sound, and other sensory inputs. Robots can learn from experience through machine learning algorithms. This capability enables them to adapt to changing conditions, improve their performance over time, and even learn from human demonstrations. They can analyze data, assess situations, and make decisions based on predefined rules or learned patterns.

Motivation behind Robotics: To cope up with increasing demands of a dynamic and competitive market, modern manufacturing methods should satisfy the following requirements:

- Reduced Production Cost
- Increased Productivity
- Improved Product Quality

Asimov's laws of robotics: The Three Laws of Robotics or Asimov's Laws are a set of rules devised by the science fiction author Isaac Asimov

1. First Law - A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. Second Law - A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.
3. Third Law - A robot must protect its own existence as long as such protection does not conflict with the First or Second Laws.

Automation and robotics:- Automation and robotics are two closely related technologies. In an industrial context, we can define automation as a technology that is concerned with the use of mechanical, electronic, and computer-based systems in the operation and control of production. Examples of this technology

include transfer lines. Mechanized assembly machines, feedback control systems (applied to industrial processes), numerically controlled machine tools, and robots. Accordingly, robotics is a form of industrial automation. Ex:- Robotics, CAD/CAM, FMS, CIMS

32.Explain the types of Automation in Robotics.

- 1) **Fixed Automation** : It is the automation in which the sequence of processing or assembly operations to be carried out is fixed by the equipment configuration. In fixed automation, the sequence of operations (which are simple) are integrated in a piece of equipment. Therefore, it is difficult to automate changes in the design of the product. It is used where high volume of production is required. Production rate of fixed automation is high. In this automation, no new products are processed for a given sequence of assembly operations. It is the automation in which the sequence of processing or assembly operations to be carried out is fixed by the equipment configuration. In fixed automation, the sequence of operations (which are simple) are integrated in a piece of equipment. Therefore, it is difficult to automate changes in the design of the product. It is used where high volume of production is required. Production rate of fixed automation is high. In this automation, no new products are processed for a given sequence of assembly operations.
- 2) **Programmable Automation** : It is the automation in which the equipment is designed to accommodate various product configurations in order to change the sequence of operations or assembly operations by means of control program. Different types of programs can be loaded into the equipment to produce products with new configurations (i.e., new products). It is employed for batch production of low and medium volumes. For each new batch of different configured product, a new control program corresponding to the new product is loaded into the equipment. This automation is relatively economic for small batches of the product.:
- 3) **Flexible Automation**: A computer integrated manufacturing system which is an extension of programmable automation is referred as flexible automation. It is developed to minimize the time loss between the changeover of the batch

production from one product to another while reloading. The program to produce new products and changing the physical setup i.e., it produces different products with no loss of time. This automation is more flexible in interconnecting work stations with material handling and storage system.

33.Explain the different types of Robots

1. Androids

Androids are robots that resemble humans. They are often mobile, moving around on wheels or a track drive. According to the American Society of Mechanical Engineers, these humanoid robots are used in areas such as caregiving and personal assistance, search and rescue, space exploration and research, entertainment and education, public relations and healthcare, and manufacturing.

2. Telechir

A telechir is a complex robot that is remotely controlled by a human operator for a telepresence system. It gives that individual the sense of being on location in a remote, dangerous or alien environment, and enables them to interact with it since the telechir continuously provides sensory feedback.

3. Telepresence robot

A telepresence robot simulates the experience -- and some capabilities -- of being physically present at a location. It combines remote monitoring and control via telemetry sent over radio, wires or optical fibers, and enables remote business consultations, healthcare, home monitoring, childcare and more.

4. Industrial robot

The IFR (International Federation of Robotics) defines an industrial robot as an "automatically controlled, reprogrammable multipurpose manipulator programmable in three or more axes." Users can adapt these robots to different applications as well. Combining these robots with AI has helped businesses move them beyond simple automation to higher-level and more complex tasks.

5.Swarm robot

Swarm robots (aka insect robots) work in fleets ranging from a few to thousands, all under the supervision of a single controller. These robots are analogous to insect colonies, in that they exhibit simple behaviors individually, but demonstrate behaviors that are more sophisticated with an ability to carry out complex tasks in total.

6. **Smart robot**

This is the most advanced kind of robot. The smart robot has a built-in AI system that learns from its environment and experiences to build knowledge and enhance capabilities to continuously improve. A smart robot can collaborate with humans and help solve problems in areas like the following:

- agricultural labor shortages;
- food waste;
- study of marine ecosystems;